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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (R,S) (FT/WP/PT) / (S2 PT) Exam April 2025 (2019 Scheme)

Course Code: MAT204

Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|---|---|---|
| 1 | Given that $f(x) = \frac{k}{2^x}$ is a probability mass function of a random variable that can take values $x = 0, 1, 2, 3$ and 4, Find (a) k (b) $P(X \leq 2)$. | 3 |
| 2 | The joint PMF of two random variables X and Y is given by $p(x, y) = c(x + y)$, $x = 0, 1, 2$; $y = 0, 1, 2, 3$. Find c | 3 |
| 3 | If X has uniform distribution in $(-a, a)$, find a such that $P(X \leq 1) = P(X > 1)$. | 3 |
| 4 | The lifetime (in years) of an electronic component is an exponential random variable with mean 1 year. Find the lifetime L which a typical component is 60% certain to exceed. | 3 |
| 5 | Find the power spectral density of the WSS power process with autocorrelation function $2e^{- \tau } + 4e^{-2 \tau }$. | 3 |
| 6 | Find the mean function of the sine wave random process with random amplitude defined by $X(t) = A \sin(\omega_0 t)$ where ω_0 is a constant and A is a random variable uniformly distributed in $(0, 1)$. | 3 |
| 7 | Evaluate $\int_0^{\pi/2} \cos x \, dx$ using trapezoidal rule in 6 sub intervals. | 3 |
| 8 | Find an approximate root of the equation $x^3 - 2x - 5 = 0$ using Regula Falsi method. | 3 |
| 9 | Using Euler's method, find y at $x = 0.25$, given $y' = 2xy$, $y(0) = 1$, $h =$ | 3 |

0.25.

- 10 Write the normal equations for fitting a curve of the form $y = a + bx$ to a given set of data points. 3

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) A random variable X has the following probability distribution 7

X	-2	-1	0	1	2	3
Y	$\frac{1}{10}$	$15k^2$	$\frac{1}{5}$	$2k$	$\frac{3}{10}$	$3k$

Find (a) k (b) $P(X < 2)$ (c) $P(X - 2 < 2)$ (d) Mean and variance of X.

- b) Derive the mean and variance of Binomial distribution. 7
- 12 a) In a binomial distribution consisting of 6 independent trials, probabilities of 1 and 2 successes are 0.28336 and .0506 respectively. Find the mean and variance of the distribution. 7
- b) Accidents occur at an intersection at a Poisson rate of 2 per day. What is the probability that in January there would be at least 3 days(not necessarily consecutive) without any accidents? 7

Module -2

- 13 a) The probability density function of a random variable X is given by 7
- $$f(x) = \begin{cases} kx^3, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$
- Find (a) $P(\frac{1}{4} < X < \frac{3}{4})$ (b) $P(X > \frac{1}{2})$.
- b) The marks obtained by a batch of students in a certain subject are normally distributed. 10% of students got less than 45 marks while 5% of students got more than 75. Find the percentage of students with score between 45 and 60. 7
- 14 a) The amount of time that a surveillance camera will run without having to be reset 7

is a random variable having exponential distribution with mean 50 days. Find the probability that such a camera will (a) have to be reset in less than 20 days (b) not have to be reset in at least 60 days.

- b) The joint density function of 2 continuous random variables X and Y is 7
- $$f(x, y) = \begin{cases} cxy, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$
- a) Find the value of c .
- b) Find $P(X \geq 3, Y \leq 4)$.
- c) Find the marginal densities of X and Y.

Module -3

- 15 a) A random process is defined by $X(t) = 2\cos(5t + \theta)$ where θ is uniformly distributed in $[0, 2\pi]$. Find the mean, autocorrelation and auto-covariance. 7
- b) If people arrive at a book stall in accordance with a Poisson process with a mean rate of 3 per minute, find the probability that the interval between 2 consecutive arrivals is (a) more than 1 minute (b) between 1 minute and 2 minutes (c) 4 minutes or less. 7
- 16 a) A cooling system in a machine fails with mean rate of 1 per week following a Poisson process. Find the probability that 2 weeks have elapsed since last failure. If we have 5 extra cooling systems and the next supply is not due in 10 weeks, find the probability that the machine will not be out of order in the next 10 weeks. 7
- b) Consider the random process $X(t) = A \cos(\omega t) + B \sin(\omega t)$ where A and B are independent random variables with zero mean and equal variance. Show that $X(t)$ is WSS. 7

Module -4

- 17 a) Solve the equation $xe^x - 2 = 0$ by Regula Falsi method. 7
- b) Find the unique polynomial $P_3(x)$ of degree 3 or less, the graph of which passes through the points $(-1, 3)$, $(0, -4)$, $(1, 5)$ and $(2, -6)$. 7

- 18 a) Compare $\int_0^1 \frac{1}{1+x^2}$ by using (i) trapezoidal method (ii) Simpson's method with step size $h = 0.25$. Compare the results with exact value obtained by actual integration. 7
- b) Using Newton's divided difference interpolating polynomial evaluate $y(8)$ from the following data. 7

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Module -5

- 19 a) Solve the following system of equations using Gauss-Jacobi method starting with the initial approximation (0,0,0). 7
- $$20x + y - 2z = 17, \quad 3x + 20y - z = -18, \quad 2x - 3y + 20z = 25.$$
- b) Fit a straight line $y = ax + b$ to the following data 7
- | | | | | | | | | |
|---|---|---|---|---|---|---|----|----|
| x | 1 | 3 | 4 | 6 | 8 | 9 | 11 | 14 |
| y | 1 | 2 | 4 | 4 | 5 | 7 | 8 | 9 |
- 20 a) Use Euler's method to solve $\frac{dy}{dx} = x + xy + y, y(0) = 1$. Compute y at $x = 0.15$ by taking $h = 0.05$. 7
- b) Use Runge-Kutta method of fourth order to find $y(0.1)$ from $\frac{dy}{dx} = \sqrt{x + y}, y(0) = 1$ taking $h = 0.1$. 7
